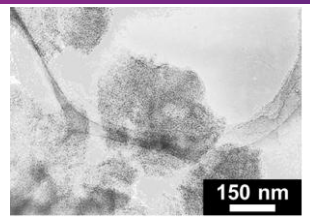
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AUSTRALIA
CREATE CHANGE

Significance and applications of nano-porous carbon materials for supercapacitor application

Assoc. Prof. Shahriar Hossain

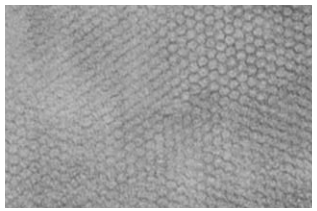
Carbon materials

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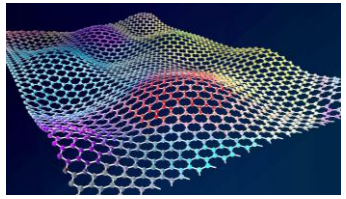
1st report of zeolite-templated porous carbon

1997



1st report of ordered mesoporous carbon

1999

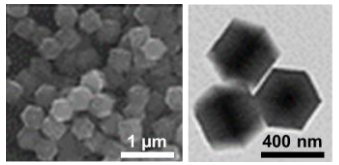


First use of the term 'Nanoarchitectonics'

2000

1st report of graphene

2004



1st report of MOF-templated carbon

2008

1st report of direct-carbonisation of MOF

2012

MOF-derived porous carbons of diverse tuneable nanoarchitectures

2020s



Future Carbon Materials

1st report of MOF

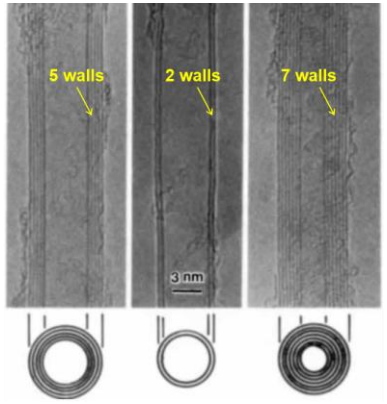
1995

1st report of MCM-41

1992

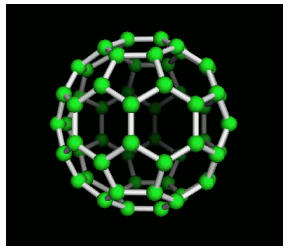
1991

1st report of CNT



1985

1st report of C_{60}

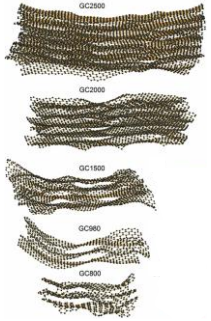


1971

1st proposal of mesoporous materials

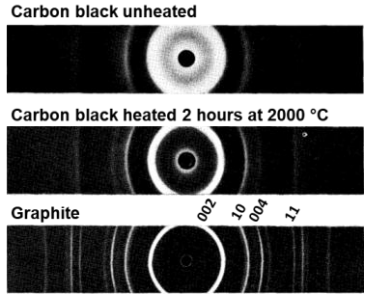
1951

Defining graphitising and non-graphitising carbons



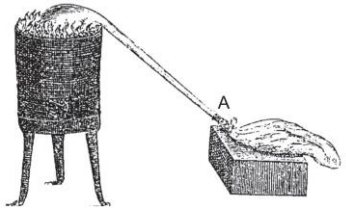
1930s

X-ray diffraction studies on carbon structure



18th century

1st scientific demonstration of gas adsorption in porous charcoal



Ancient times

Early usage of porous charcoal for various purposes

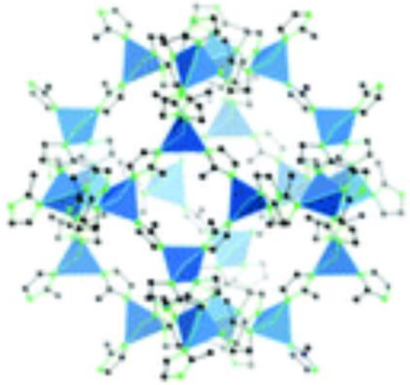


Carbon Materials in Energy Applications

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Advantages	Disadvantages
High structural strength but very light *e.g. graphene, carbon nanotube, etc.*	***Still quite expensive*** *e.g. 500 mg rGO – 621.00 AUD (sigma Aldrich)* *1 g SWCNT – 1800.00 AUD (sigma Aldrich)*
Electrically conductive	***Mass-production is not yet feasible***
Chemically inert	***Bears morphological limitations*** *e.g. serious aggregation and stacking happens on electrode*

Alternatives?



Metal-organic framework (MOF)

Carbonization



Microporous MOF-derived carbon

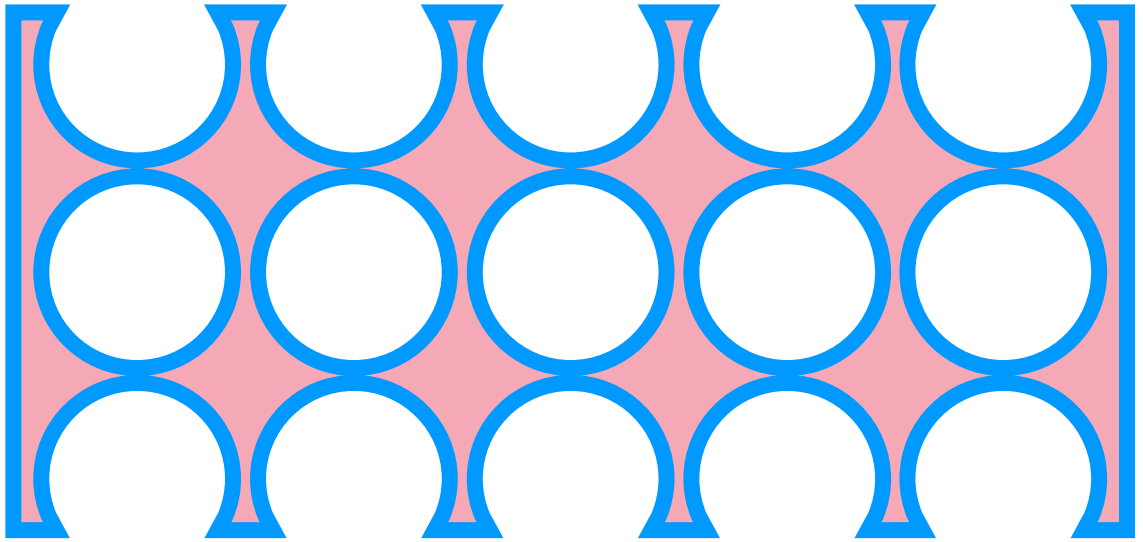
Porous Carbon Materials in Energy fields

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Carbon material containing nanopores (voids).

- High specific surface area
- Tuneable nanopore size





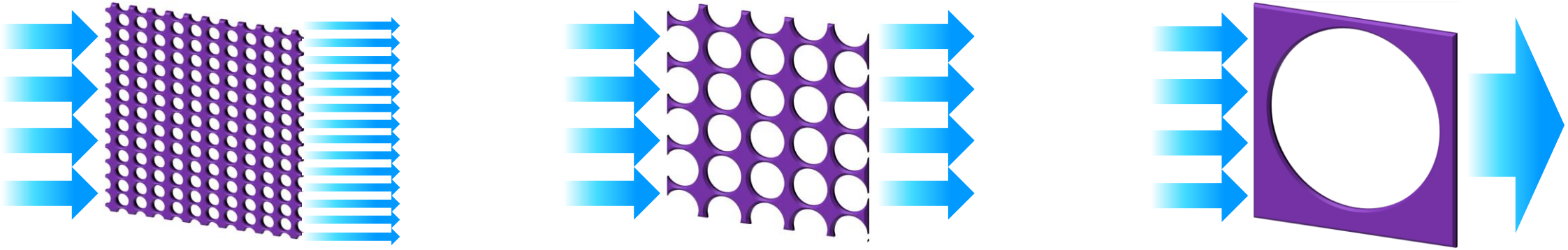
Classes of nanopores

- Micropore (diameter < 2 nm)
- Mesopore (2 nm ≤ diameter < 50 nm)
- Macropore (50 nm ≤ diameter)

Different Porosity, Different Advantage

The University Of Queensland
AUSTRALIA

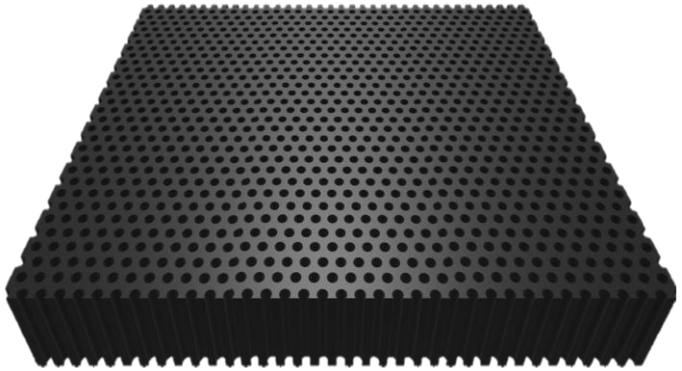
Specific surface area



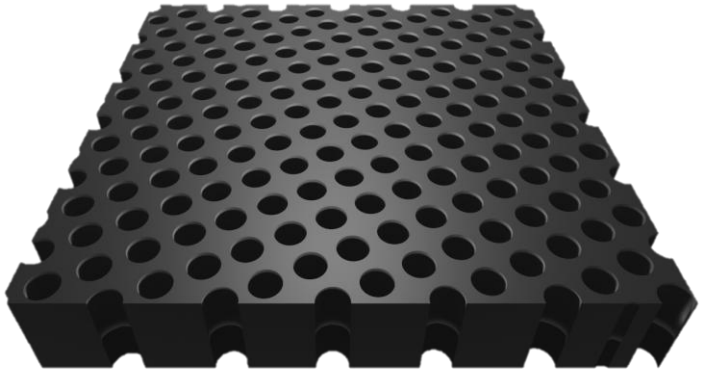
Microporous *Mesoporous* *Macroporous*

Diffusion

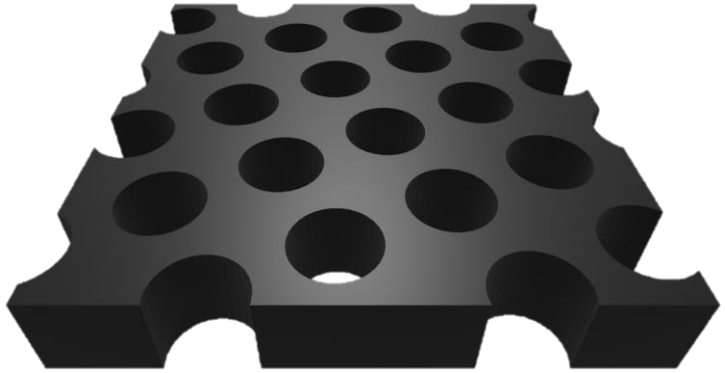
General effect of pore size



Micropore
(pore diameter < 2 nm)



Mesopore
(2 nm ≤ pore diameter < 50 nm)



Macropore
50 nm ≤ Pore diameter

Pore size ↑ → Surface area ↓ + Diffusion ↑

Pore size ↓ → Surface area ↑ + Diffusion ↓

Nanoarchiectonics

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'*Nanoarchitectonics*' refers to the concept of arranging nano-sized structural units in an intended configuration.

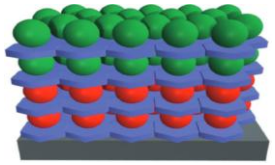
E.g.,



Porous structure

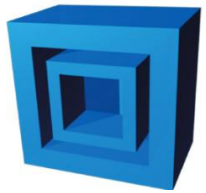


Layer-by-Layer structure



Layer-by-Layer heterostructure

$$A + B\,(+ C + D + \ldots..) = X + \alpha$$



Muti-shell structure



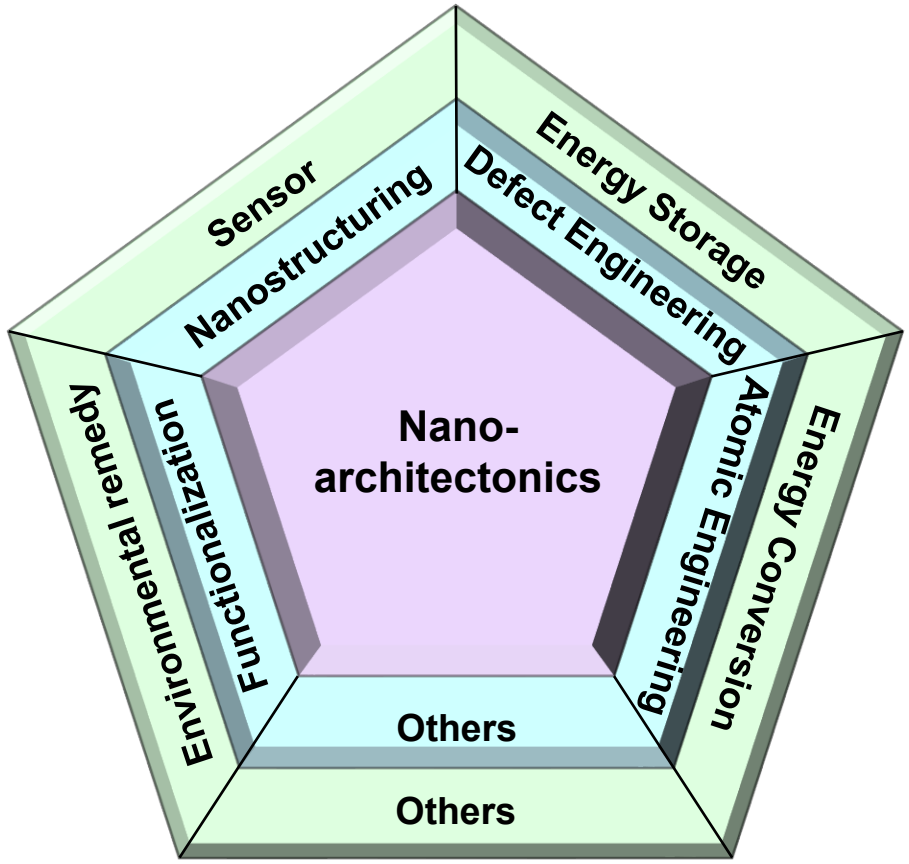
Core-shell structure



Framework structure

Why is '*Nanoarchitectonics*' worth of consideration?

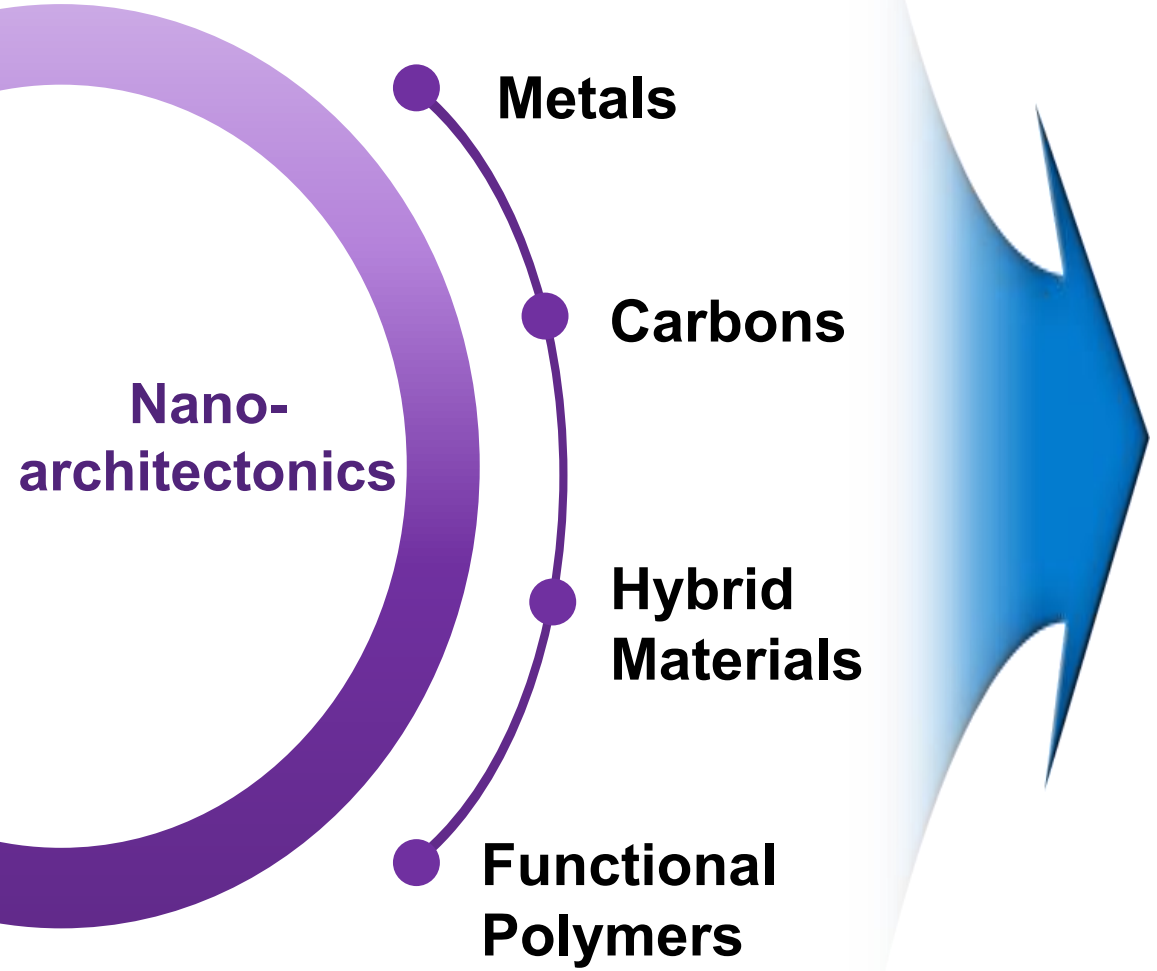
- It helps to design nanomaterials in a more efficient manner.
- It offers a wide range of synthetic methods to obtain the material of interest.
- It helps to understand the interplay between various properties of materials.

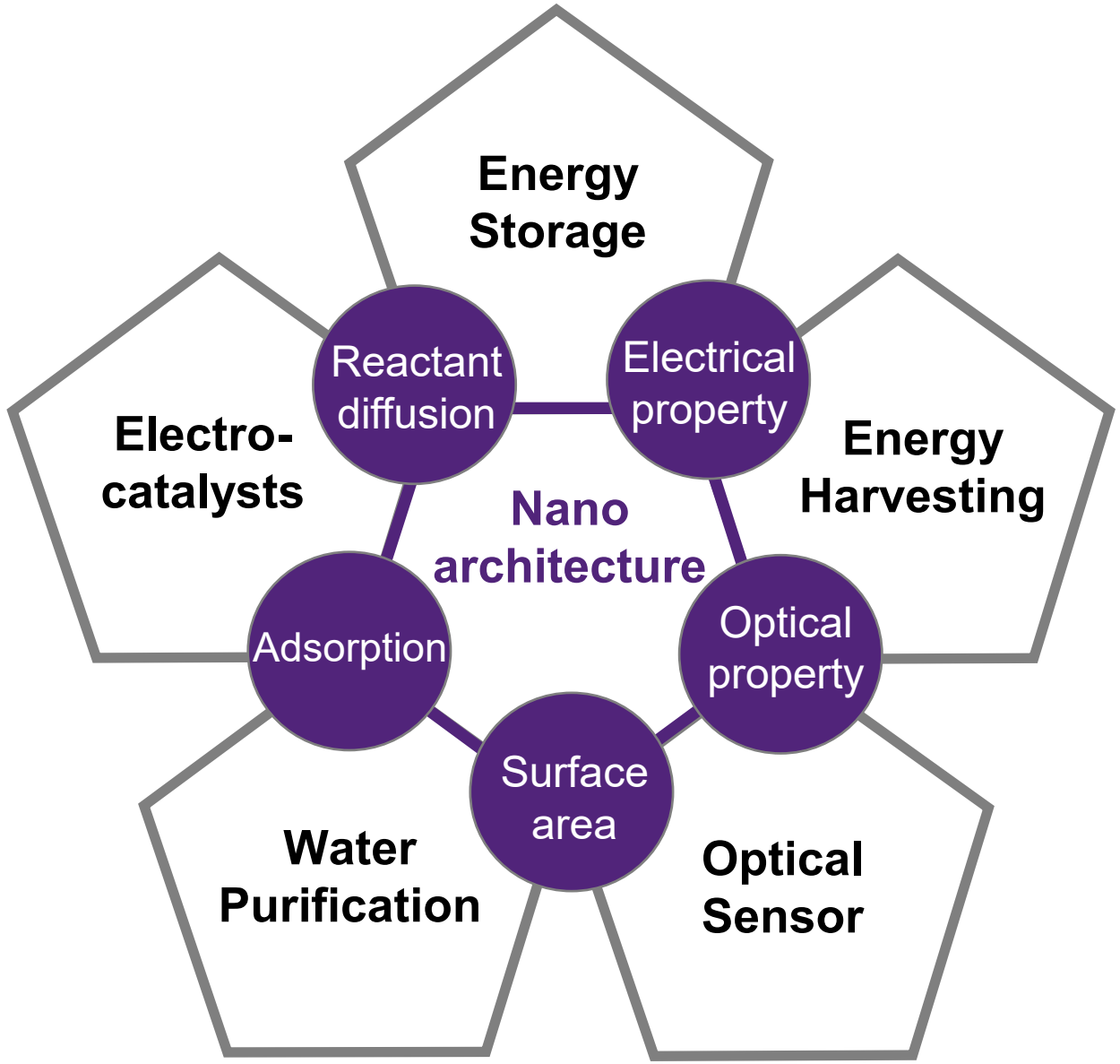


Nano-architectonics

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"Nano-architectonics"



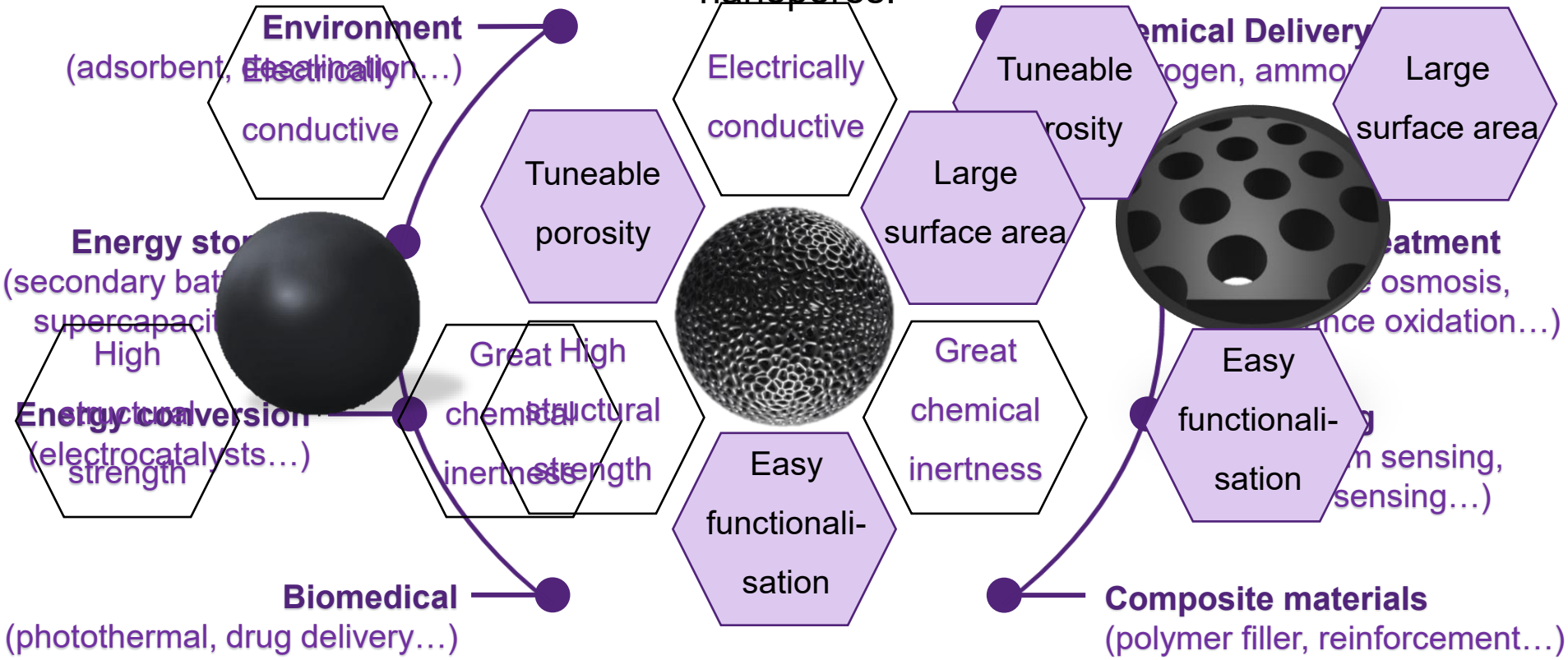


Nano-architectonics

Carbon materials are mainly composed of carbon atoms.

Porous carbon materials are mainly composed of carbon atoms and possess nanopores.

Porous materials are materials that possess pores/voids in nano-scale.

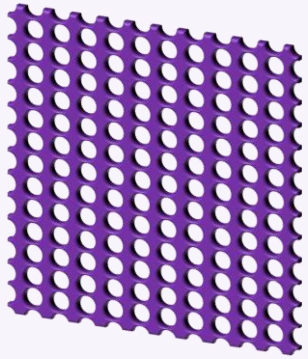


Just Porous is not enough - Nanoarchitectonics

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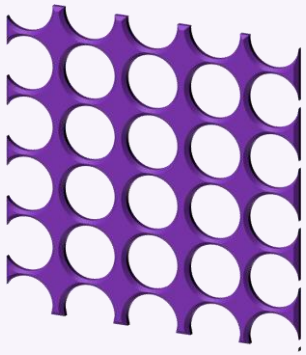
Nanoarchitectonics

➡ *Scientific study of nanoscale architectures of nanomaterials*



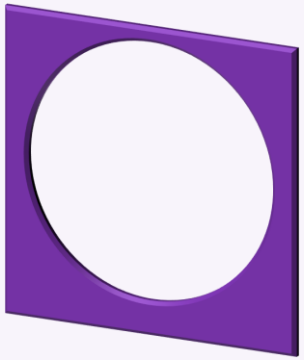
Microporous

or



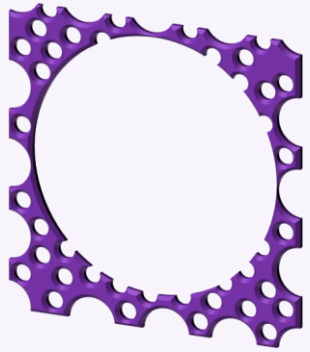
Mesoporous

or

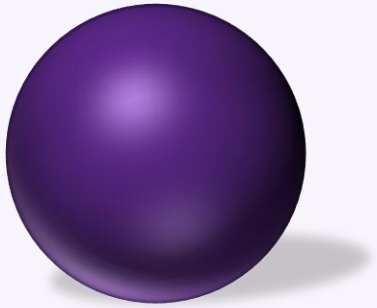


Macroporous

or

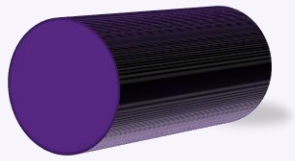


Hierarchical



0D

or



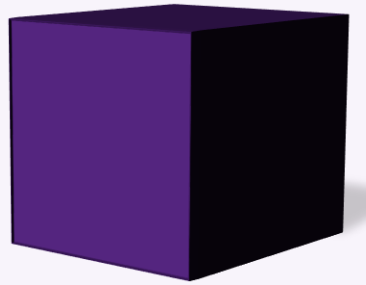
1D

or



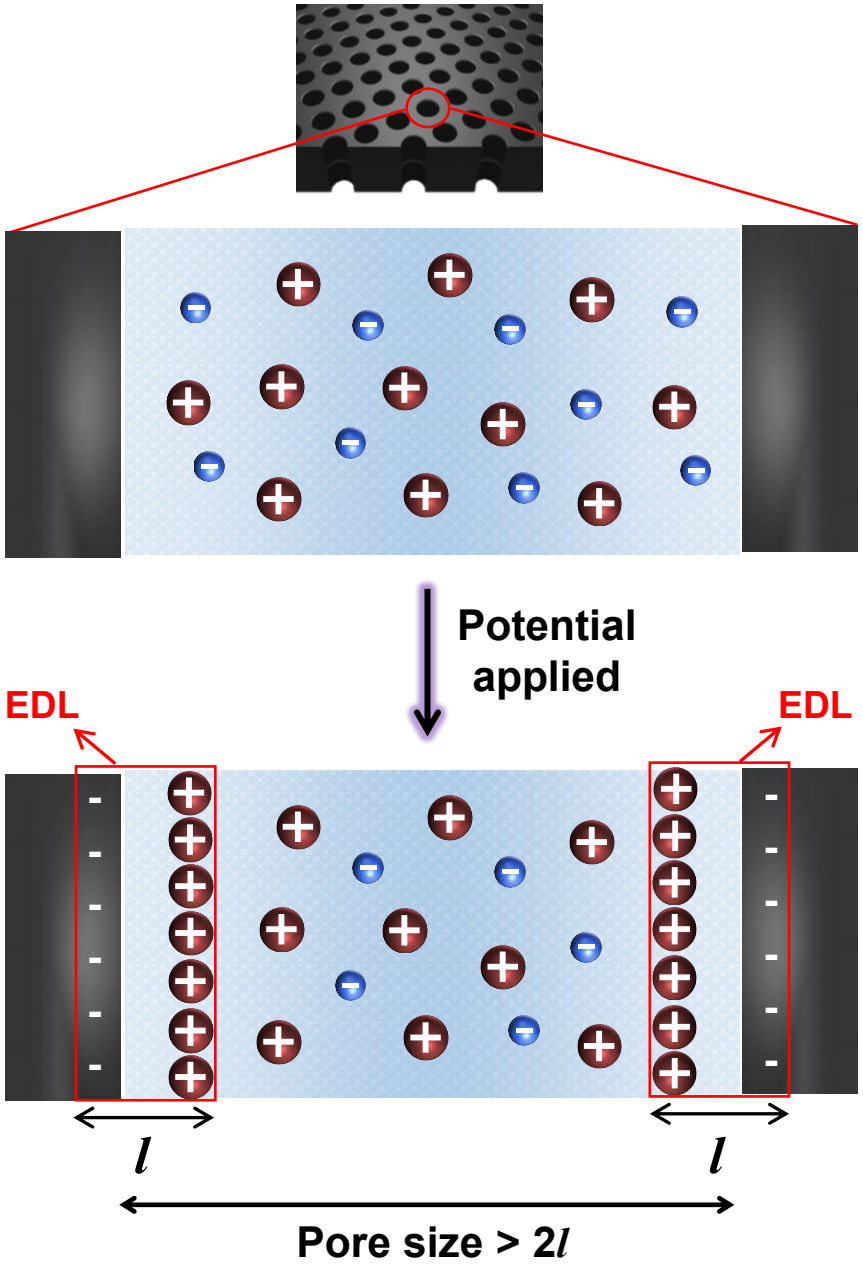
2D

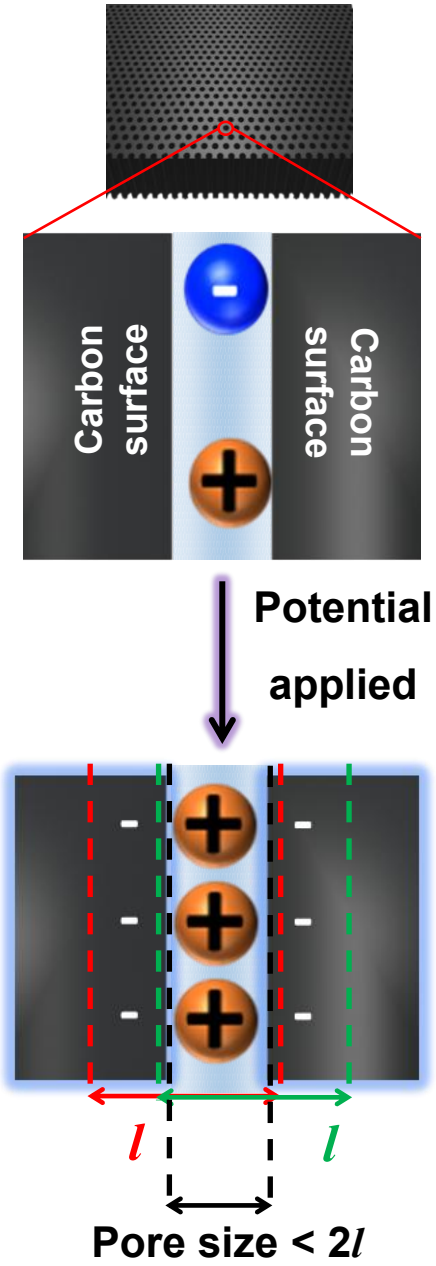
or

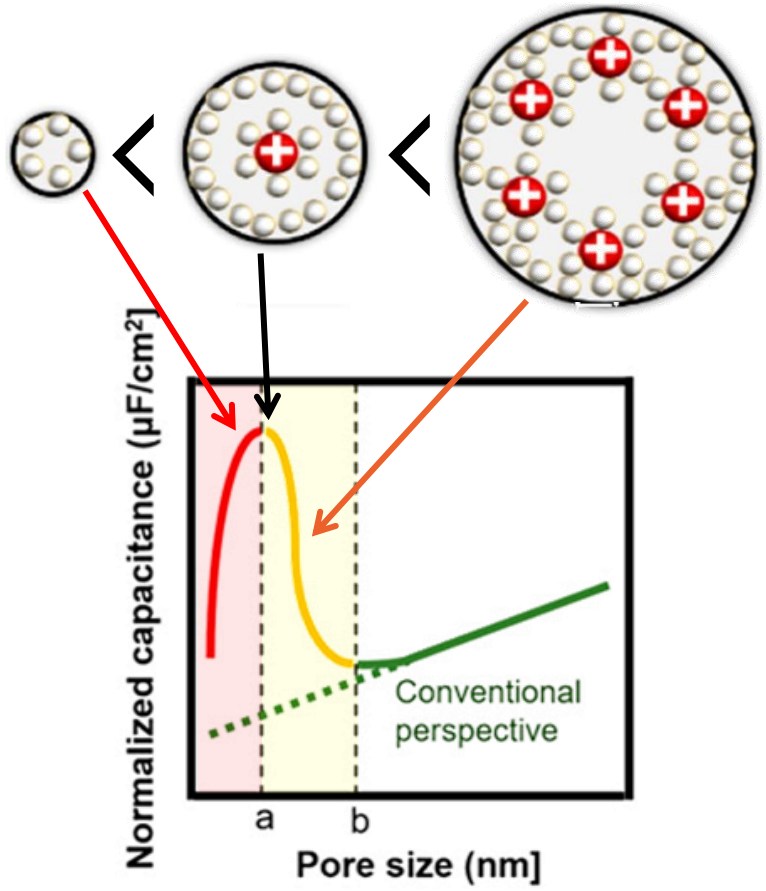


3D

Electrochemical effect of pore size

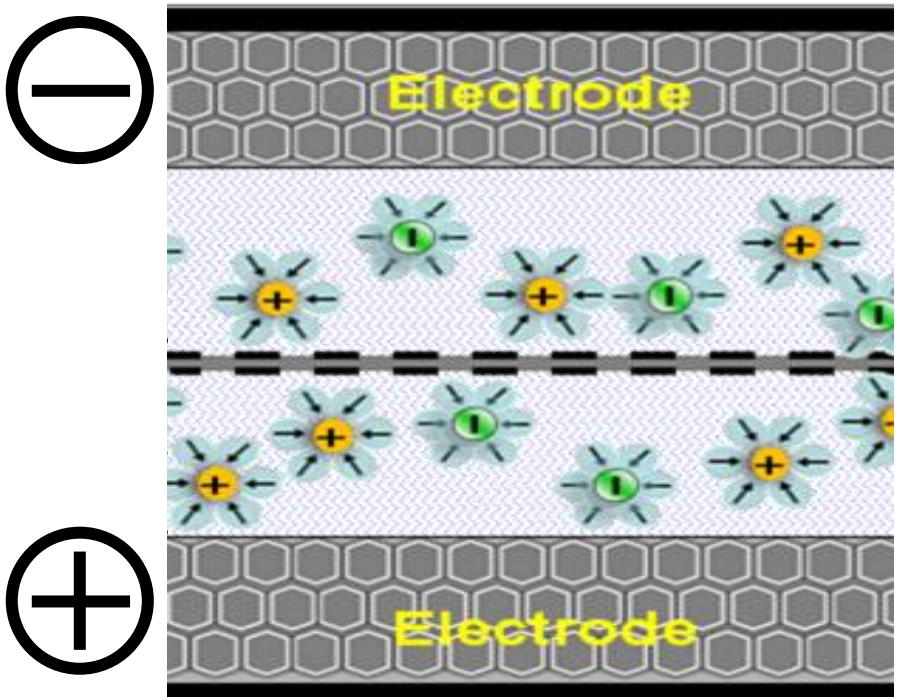






'Optimized pore size for a specific electrochemical application is necessary.'

EDLC : Electric Double Layer Capacitance



1. Potential applied
2. Electrosorption of ions &

   Polarisation of electrode
3. Charge accumulation on surface

Surface area of electrode material is important !

What are the problems with MOF

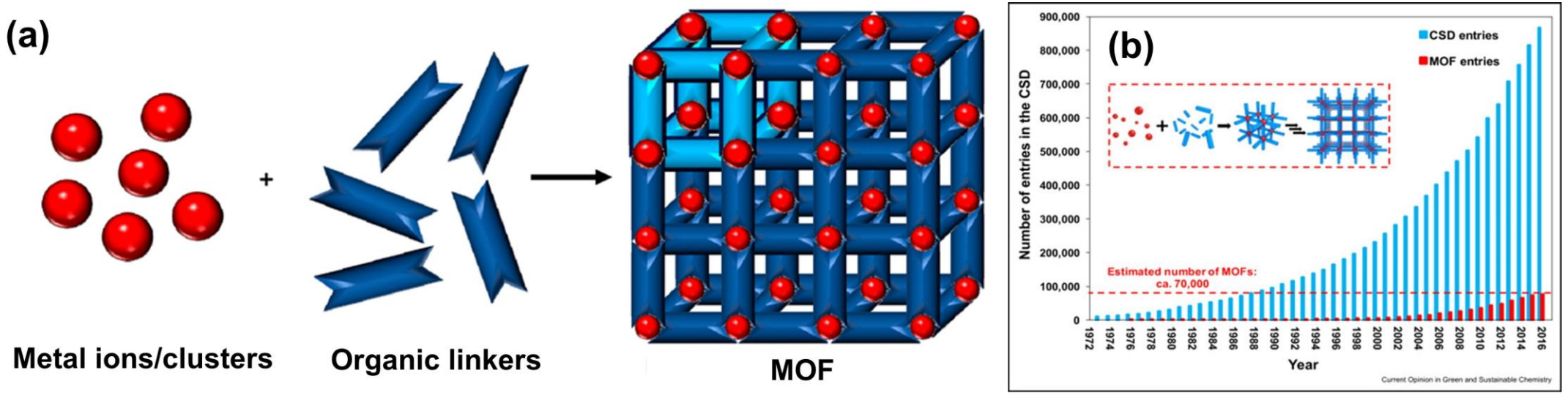


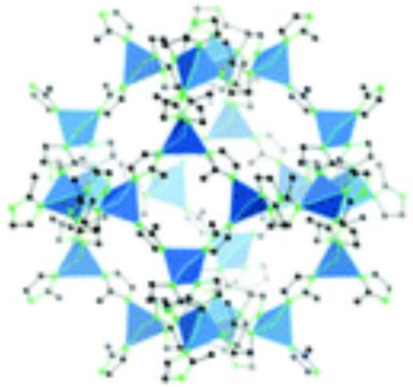
Figure 4. (a) Schematic illustration showing the general assembly process of MOF. (b) A plot showing the number of MOFs listed in Cambridge Crystal Database from 1972 to 2016.

Metal-organic frameworks (MOFs), or porous coordination polymers, are crystalline porous materials formed by coordination bonding between inorganic and organic species on the basis of the self-assembly of the reacting units.

i. Large specific surface areas,
ii. Ultrahigh porosities
iii. Excellent thermal and chemical stabilities
iv. Their great potential for chemical and structural modifications
v. Excellent candidates for versatile applications.

Poor electrical conductivities substantially limit their potential for electrochemical/energy storage applications

MOF-derived Microporous Carbon



Pyrolysis



Metal-organic framework (MOF)

- Extremely high specific surface area
- Maximize contact area
- Promising electrochemical performance

BUT

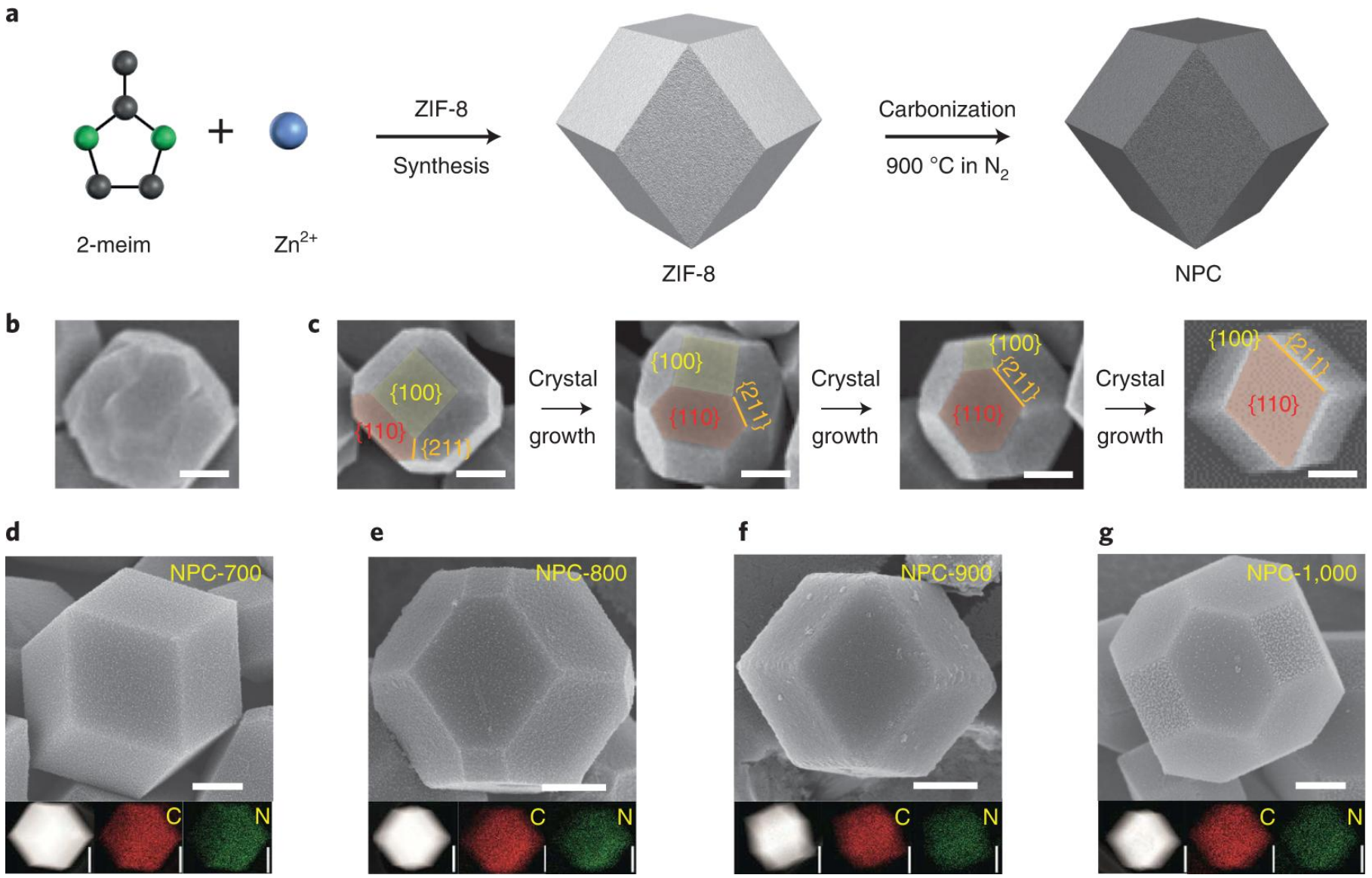
MOF-derived Nanoroporous Carbor

- Low level of graphitization
- Poor diffusion efficiency caused by microporosity
- A lack of active sites required for certain electrochemical applications

Nanoarchitectonics

- ❑ Tailored control of structures (porosity, morphology, dimensions, etc.) and
- ❑ Chemical compositions (impregnation, functionalization, doping, etc.)

Carbonisation of ZIF-8 using various temperatures?

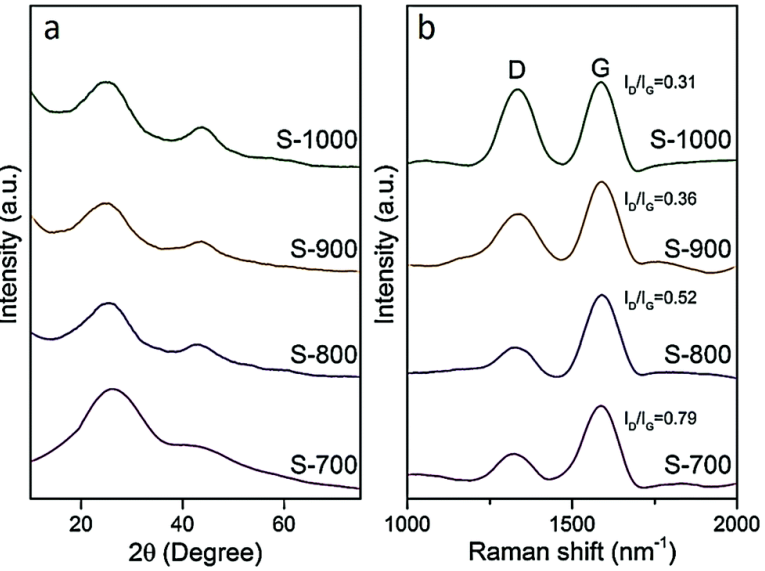


1. Nitrogen-doped carbon materials present extraordinarily high capacitance due to a substantially higher positive charge density on the carbon atoms around them

2. HF washing was used to remove the metals, this is the reason of having only wel distributed C and N without any Zn

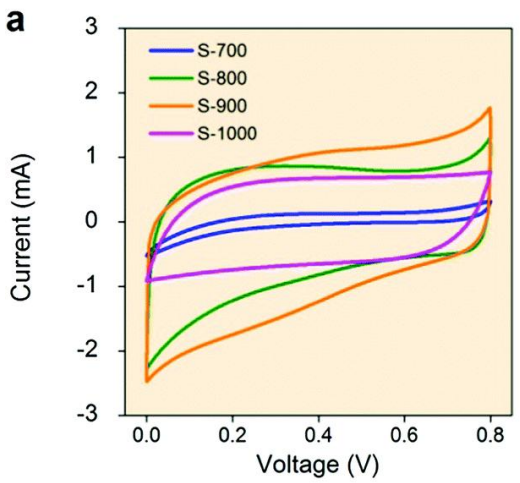
https://doi.org/10.1038/S41596-022-00718-2; Phys. Chem. Chem. Phys., 2016, 18, 29308

Carbonisation of ZIF-8 using various temperatures?



The roles of the nature of carbon on supercapacitor application –

1. **Graphitic carbon**
2. **Micro, Meso, macro**
3. **carbon, nitrogen and other impurity**



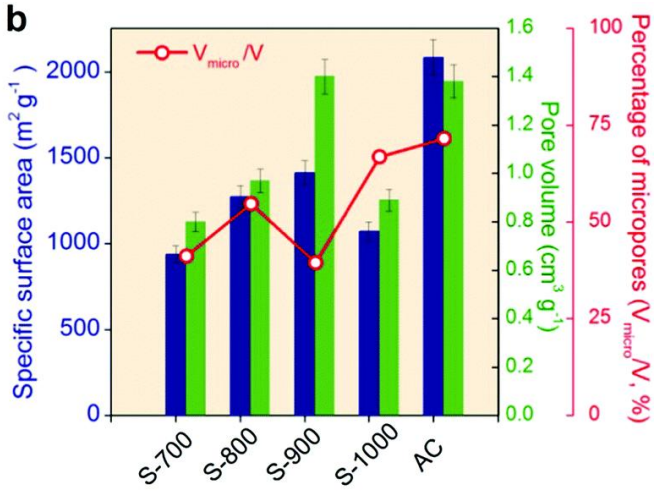
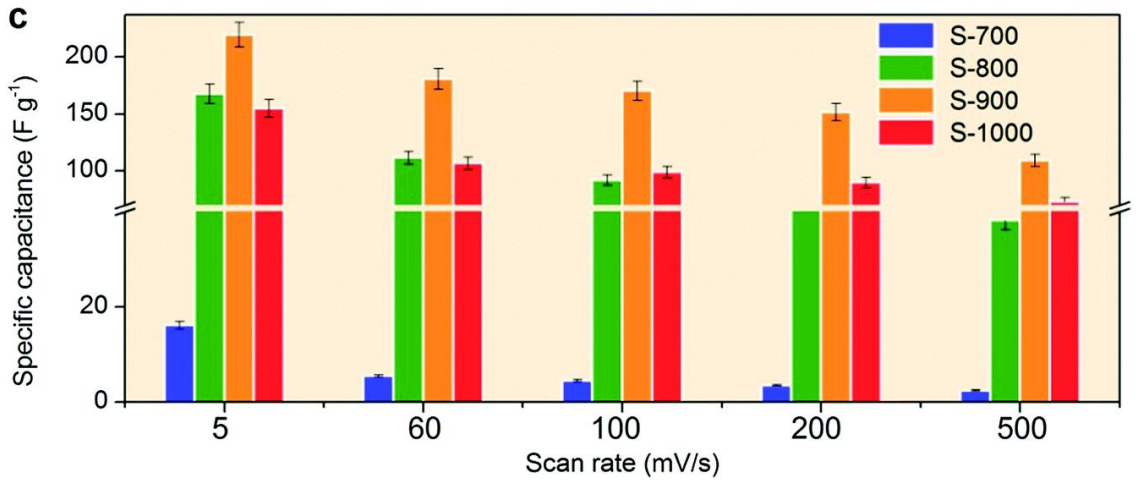


Table S1. The proportion of different element contents analyzed by XPS analysis.

Samples	Carbon (at.%)	Nitrogen (at.%)	Oxygen (at.%)	Zinc (at.%)	Other elements (at.%)
S-700	67.6	14.2	14.4	0.9	2.9
S-800	77.9	13.6	7.0	1.2	0.3
S-900	82.9	11.9	4.2	0.6	0.4
S-1000	90.3	3.5	4.3	0.1	1.8

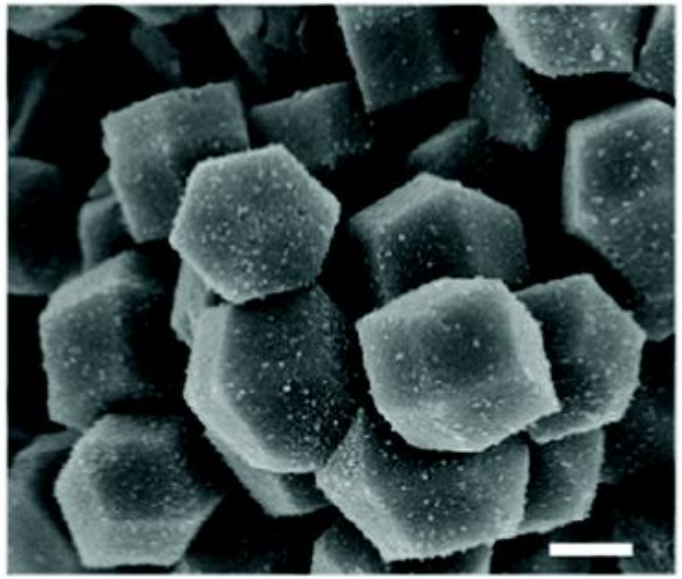


https://doi.org/10.1038/S41596-022-00718-2; Phys. Chem. Chem. Phys., 2016, 18, 29308

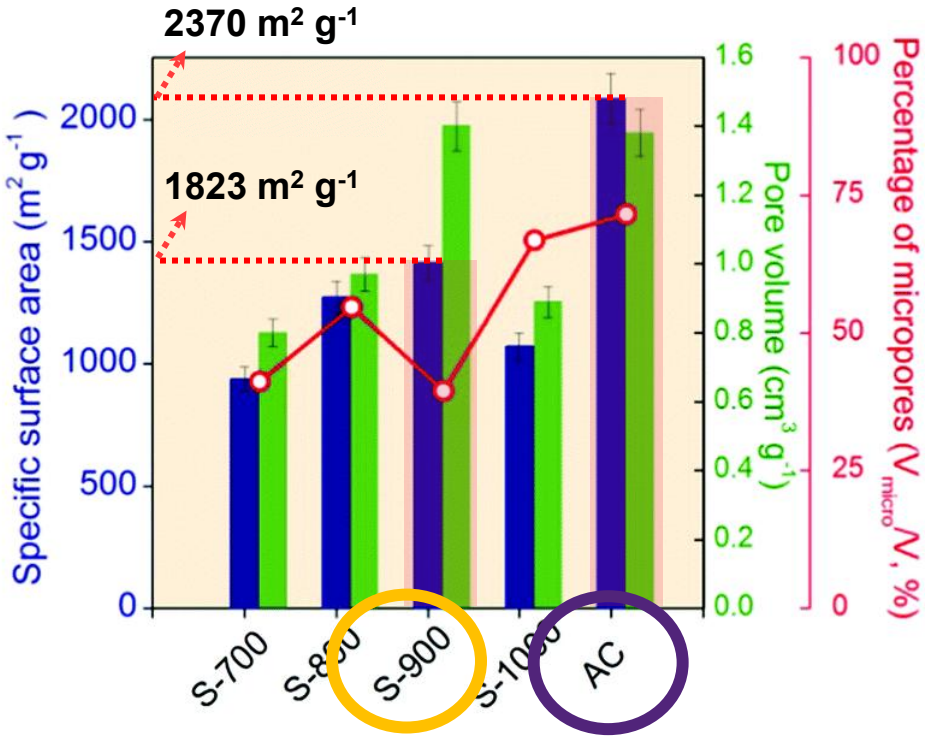
High surface area, high capacitance?

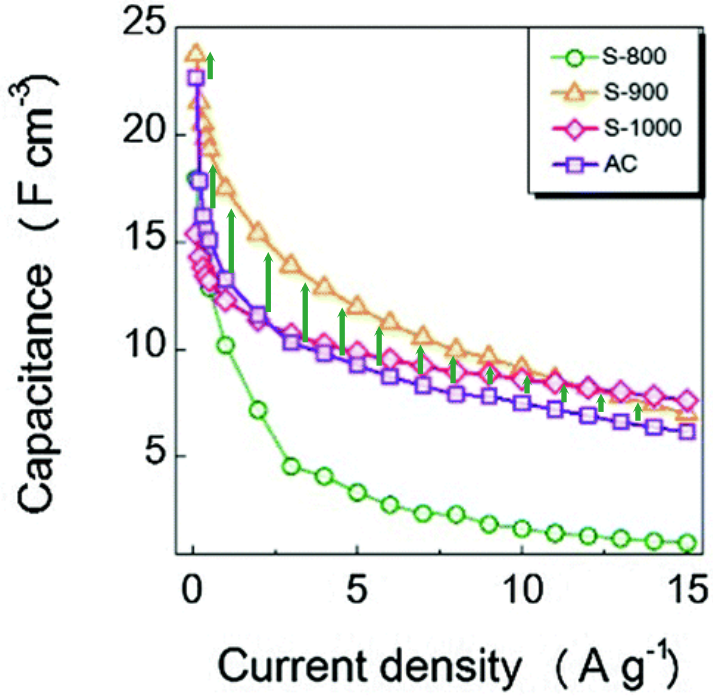


ZIF-8 ZIF-8 Carbon



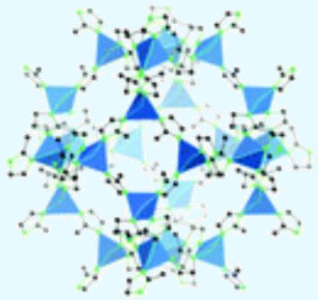
ZIF-8 Carbon





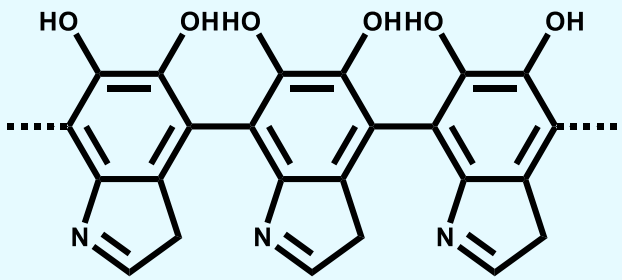
Phys. Chem. Chem. Phys., 2016, 18

Nanoarchitecture is the key to high-performance



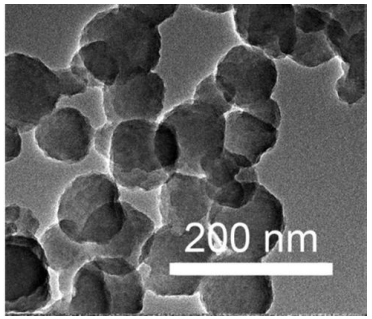
ZIF-8

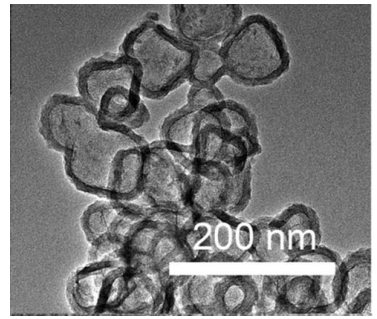
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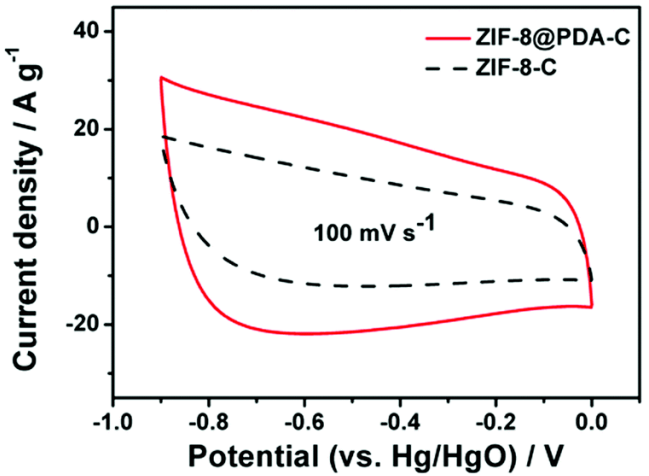
Polydopamine

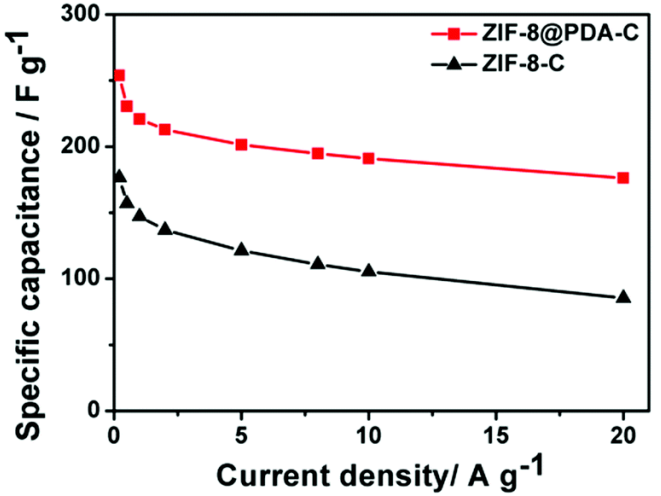






ZIF-8 Carbon	ZIF-8@PDA Carbon
~ 1000 $m^2 g^{-1}$	316 $m^2 g^{-1}$
177 F g^{-1} at 0.2 A g^{-1}	253 F g^{-1} at 0.2 A g^{-1}





New J. Chem., 2018, 43

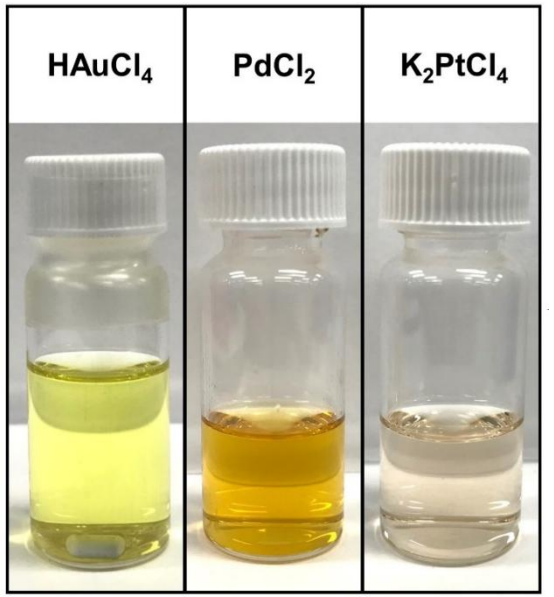
Mesoporous Metal: Soft-template Method



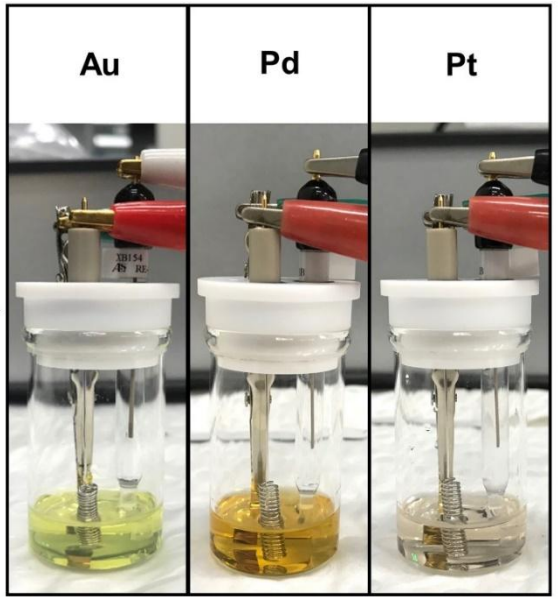
EtOH

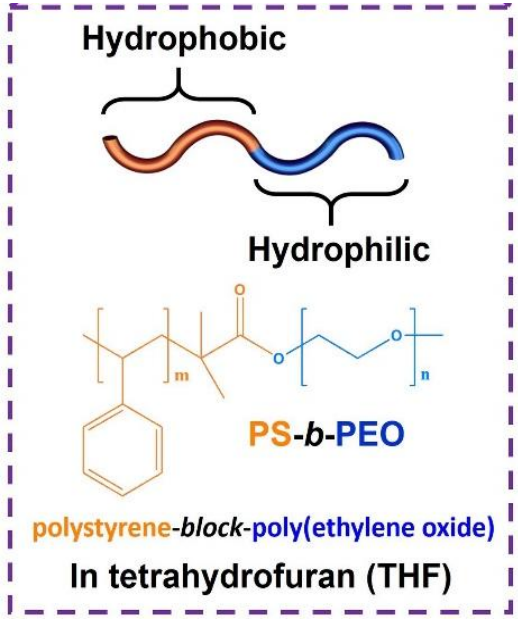
H_2O

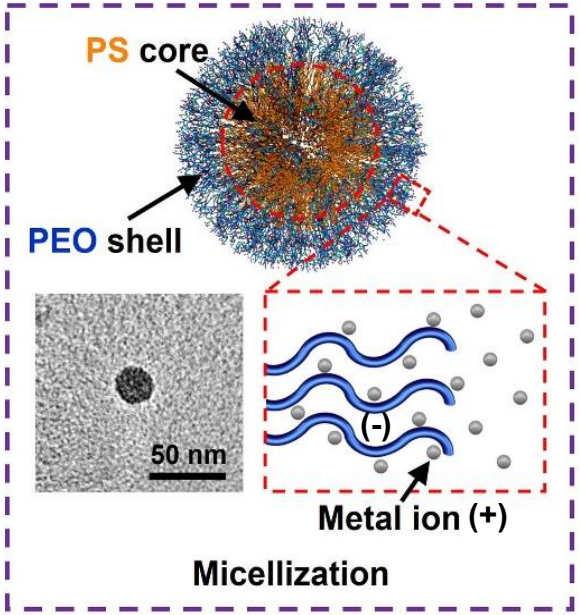
Metal precursor

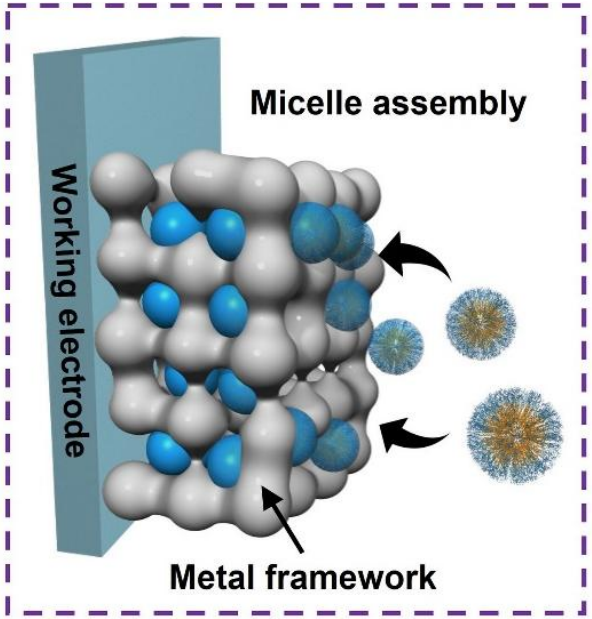


Applied Voltage





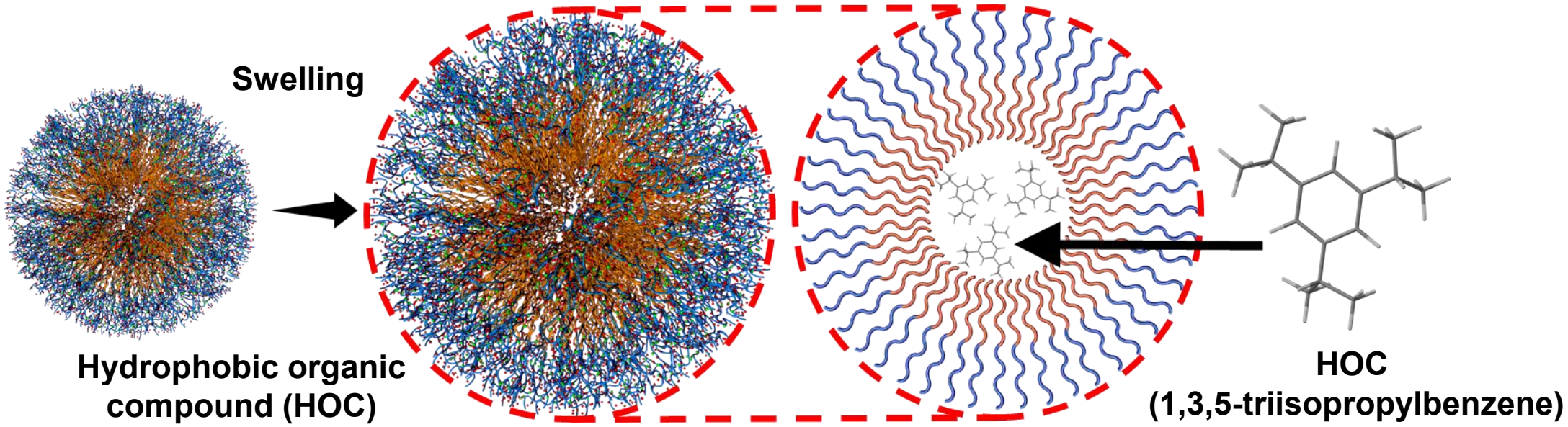


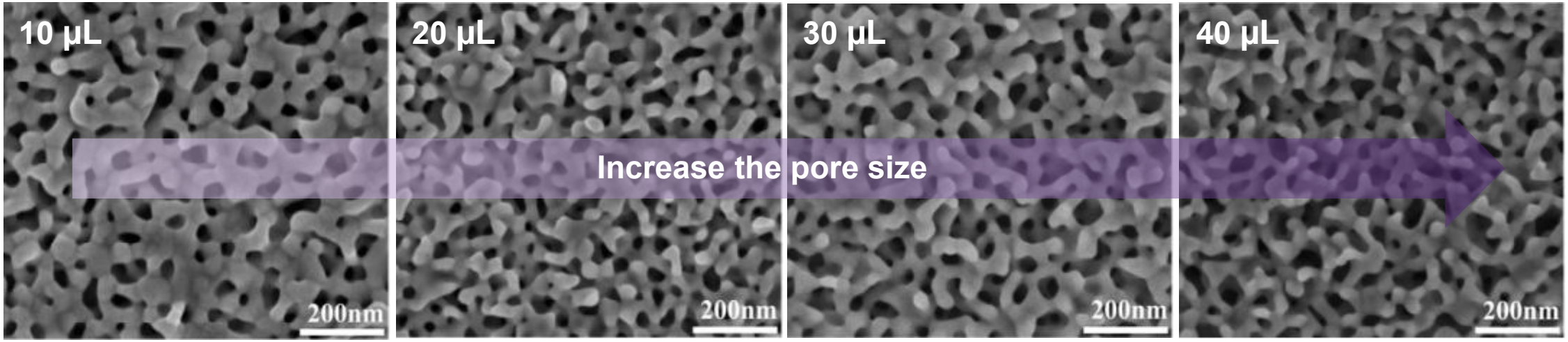


Deposition potential → Optimized through linear sweep voltammetry (LSV)

Nat. Protoc. 2020, 15, 2980-3008

Mesoporous Metal: Nanostructure Control



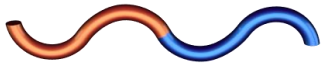


Nat. Protoc. 2020, 15, 2980-3008

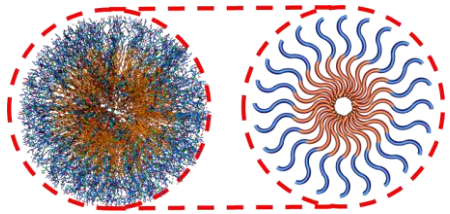
Mesoporous Metal: Nanostructure Control

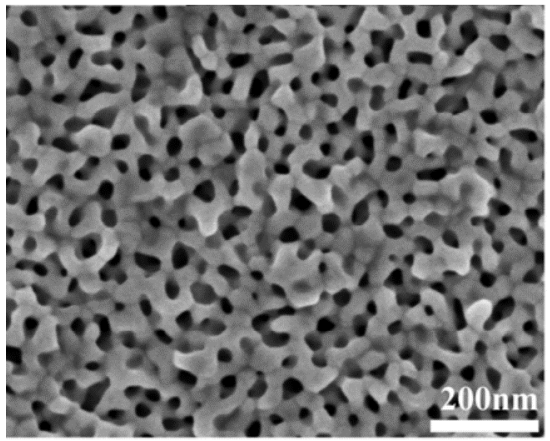
THE UNIVERSITY OF QUEENSLAND AUSTRALIA

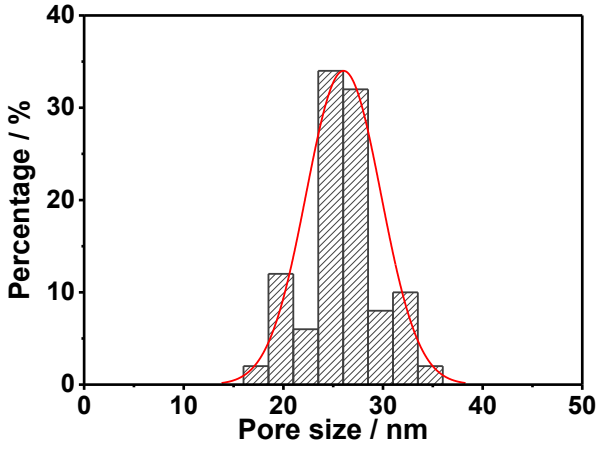
Low molecular weight



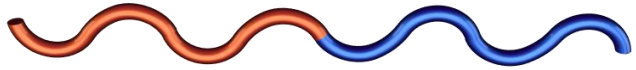
PS_{18000}-*b*-PEO_{7500}



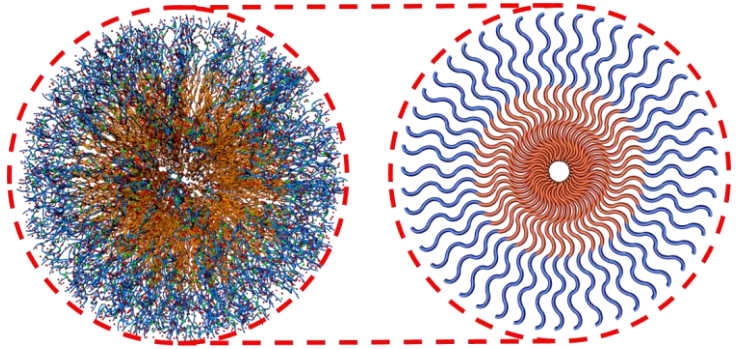


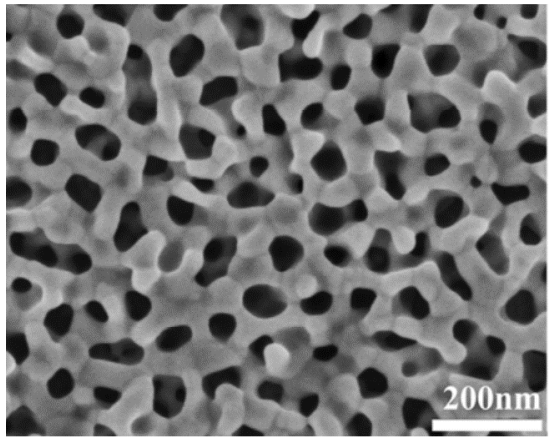


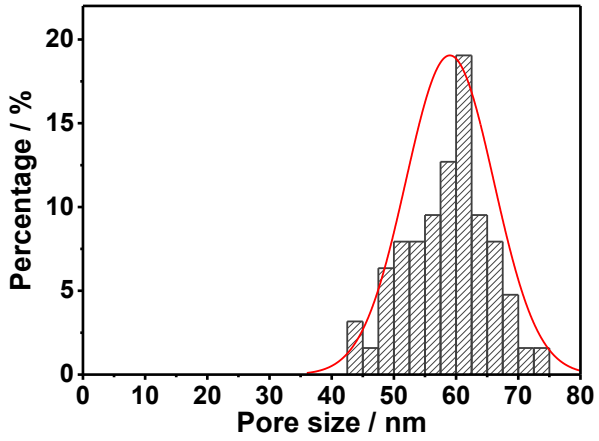
High molecular weight



PS_{63000}-*b*-PEO_{26000}



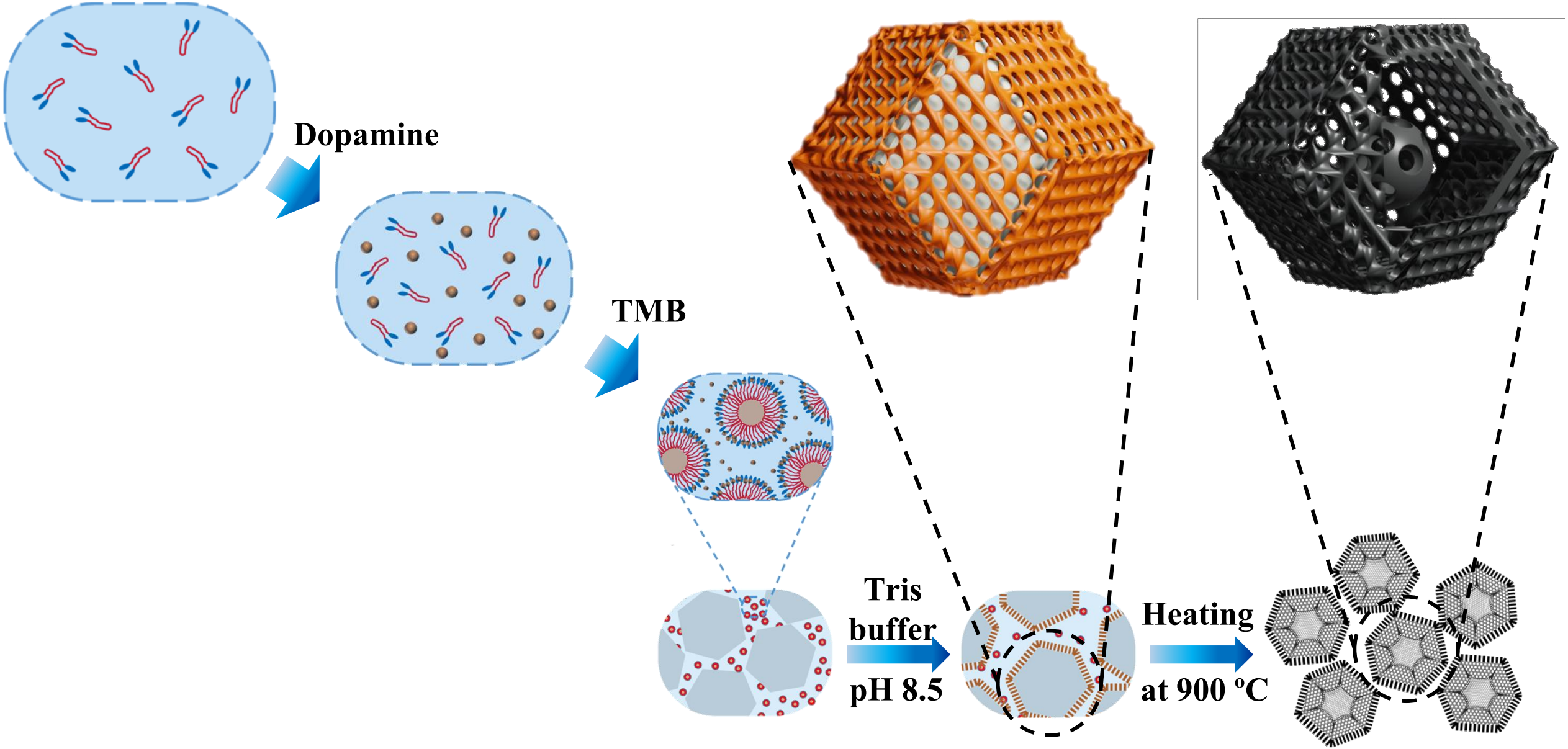




Metal pore size → control by molecular weight of polymer

Nat. Protoc. 2020, 15, 2980-3008

Mesostructured Polydopamine-coated ZIF-8

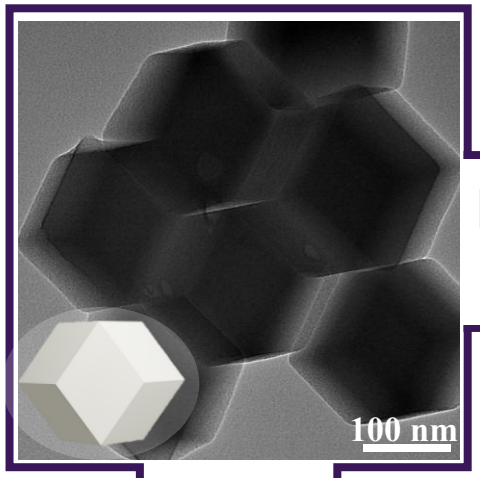


Tailored Nanoarchitecturing of Microporous ZIF-8 to Hierarchically Porous Double-Shell Carbons and Their Intrinsic Electrochemical Property | ACS Applied Materials & Interfaces

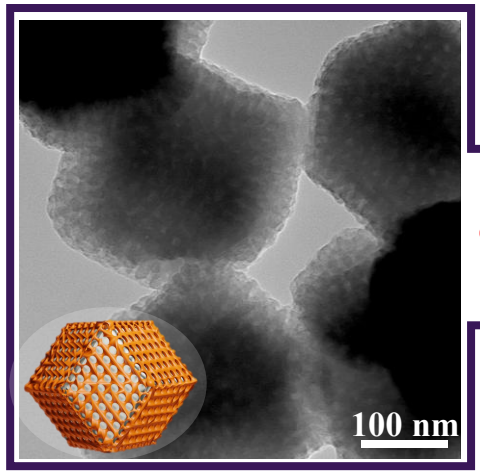
Mesostructured Polydopamine-coated ZIF-8

THE UNIVERSITY OF QUEENSLAND AUSTRALIA

Route II: *Mesostructured PDA coating & Pyrolysis*



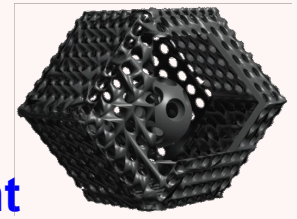
Polydopamine with micelle



900 °C

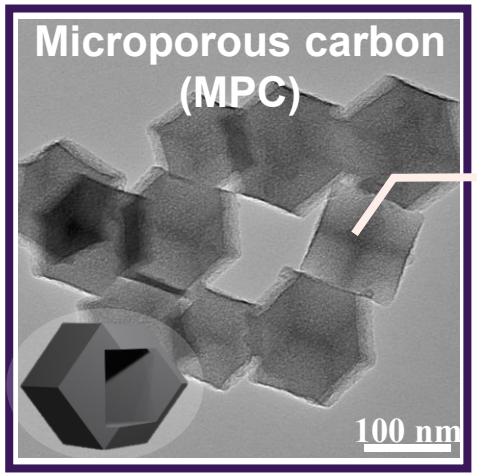


- Micro-, Meso & Macroporous
- Core cavity
- Diffusion efficient



Route I: *Direct pyrolysis*

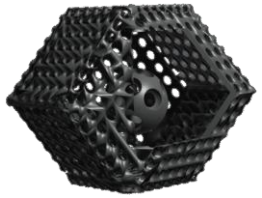
900 °C

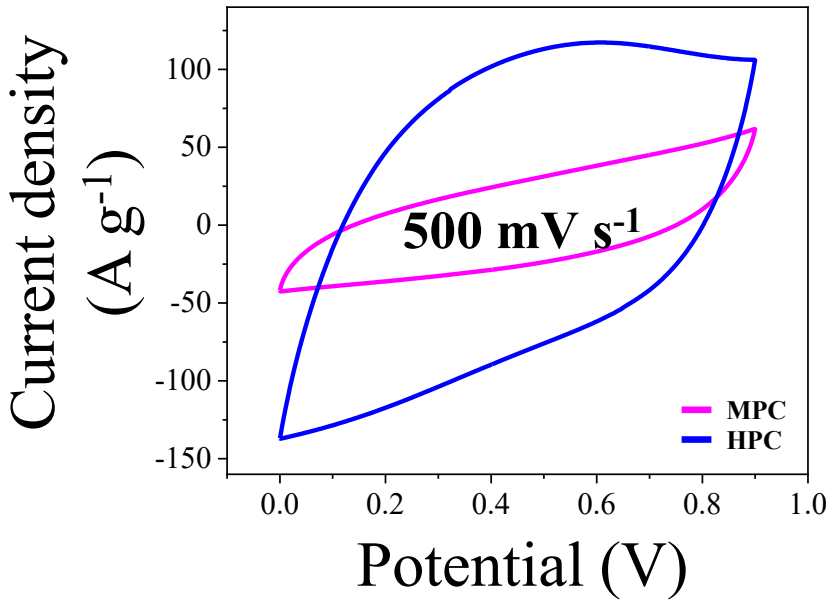


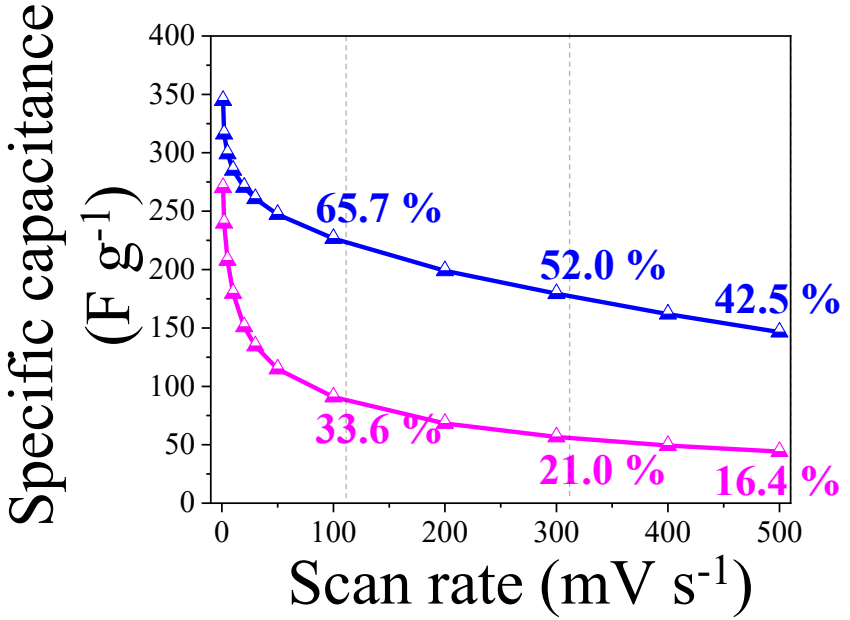
- Microporous
- Dense core
- Diffusion restriction

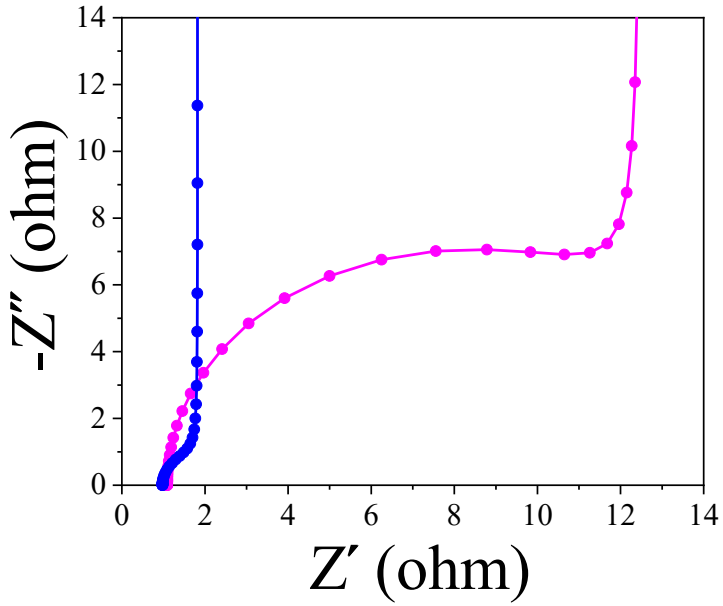


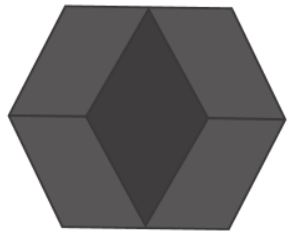
Electrochemical analysis for Supercapacitor

		Specific surface area	Specific capacitance	
MPC		993 $m^2 g^{-1}$	270 F g-1 (1 mV s-1)	44.2 F g^{-1} (500 mV s^{-1})
HPC		563 $m^2 g^{-1}$	345 F g^{-1} (1 mV s^{-1})	147 F g^{-1} (500 mV s^{-1})







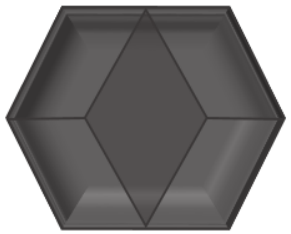


Microporous Carbon

☺ High surface area

☹ Poor diffusion



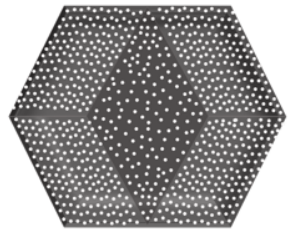


Hollow Carbon

☹ Low surface area

☺ Good diffusion





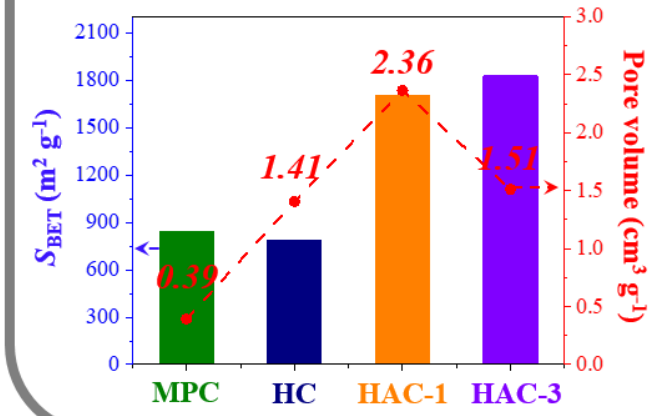
Hollow Activated Carbon

☺ High surface area

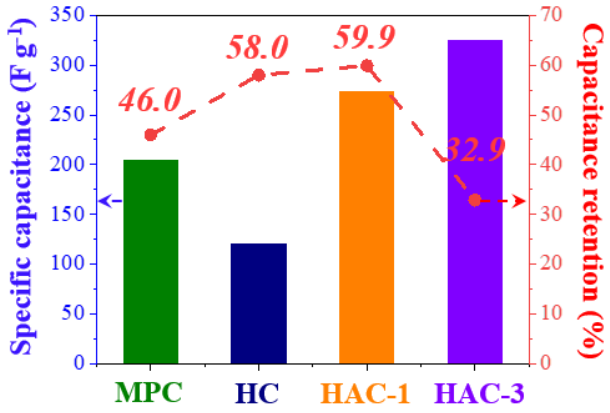
☺ Good diffusion



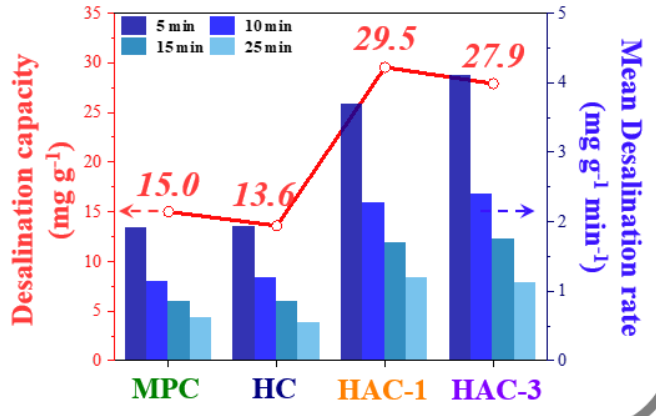
BET Analysis



Supercapacitor



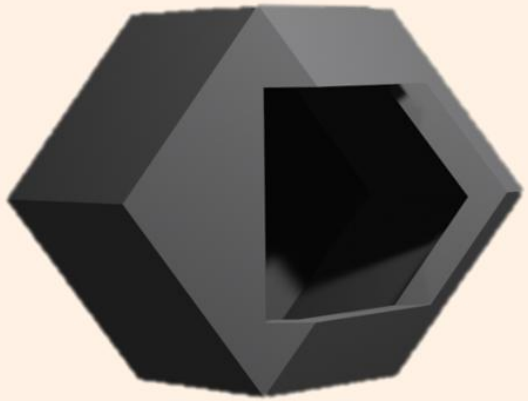
Capacitive deionization

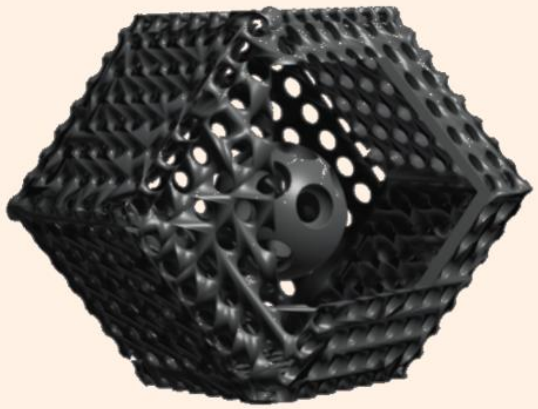


Electrochemical analysis for Supercapacitor

The University Of Queensland
AUSTRALIA







Microporous Carbon

- ✓ Microporous
- ✓ Diffusion highly restricted
- ✓ Active sites largely inaccessible

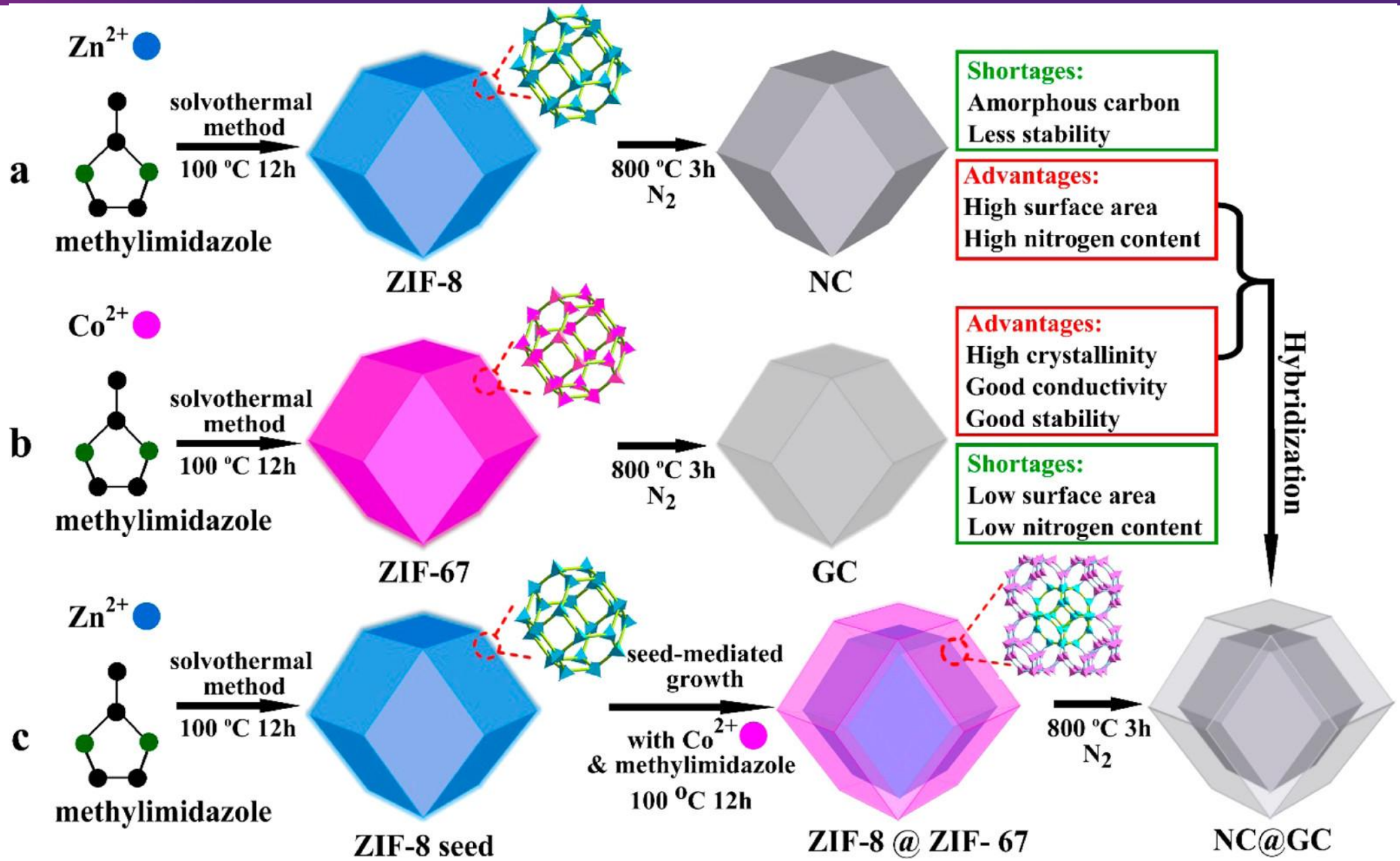
Hollow Microporous Carbon

- ✓ Micro- & Macroporous
- ✓ Diffusion is efficient
- ✓ Active sites moderately accessible

Hierarchically Porous Carbon

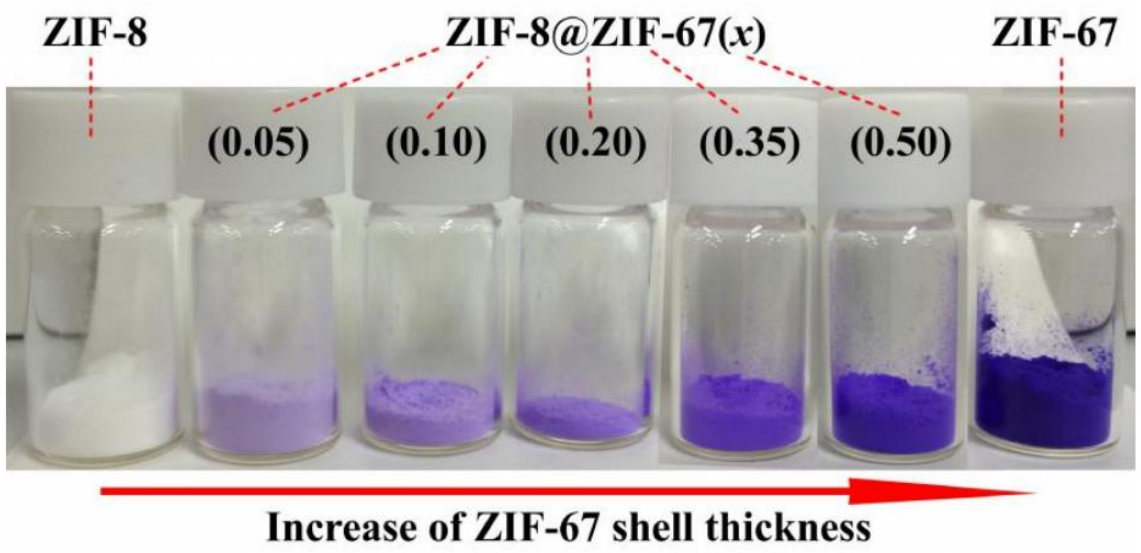
- ✓ Micro-, Meso- & Macroporous
- ✓ Diffusion is efficient
- ✓ Active sites highly accessible

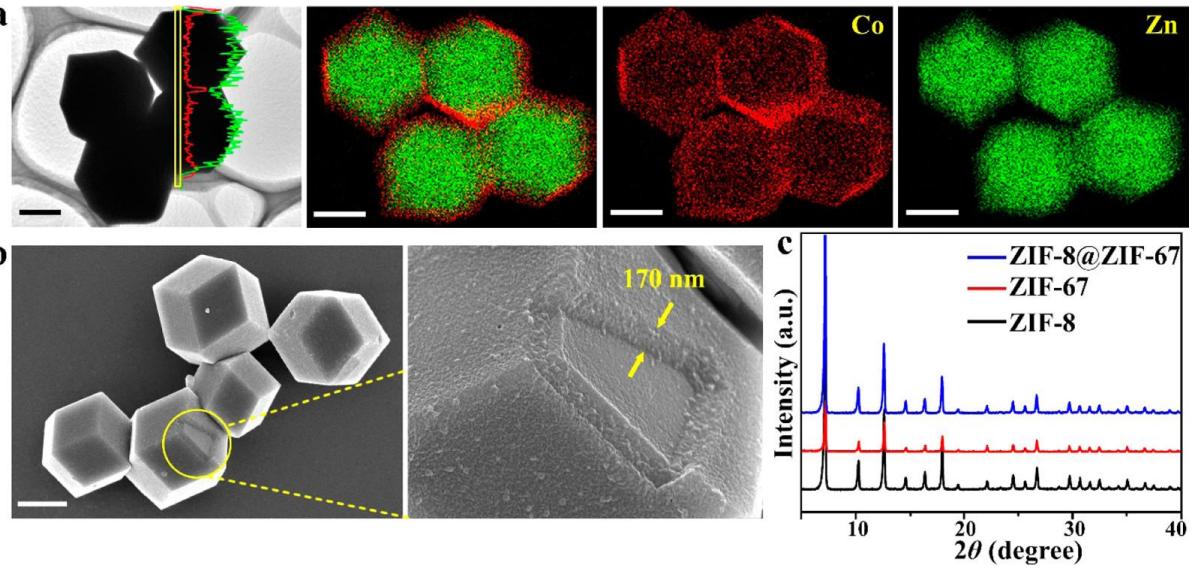
Core-shell MOF for Supercapacitor

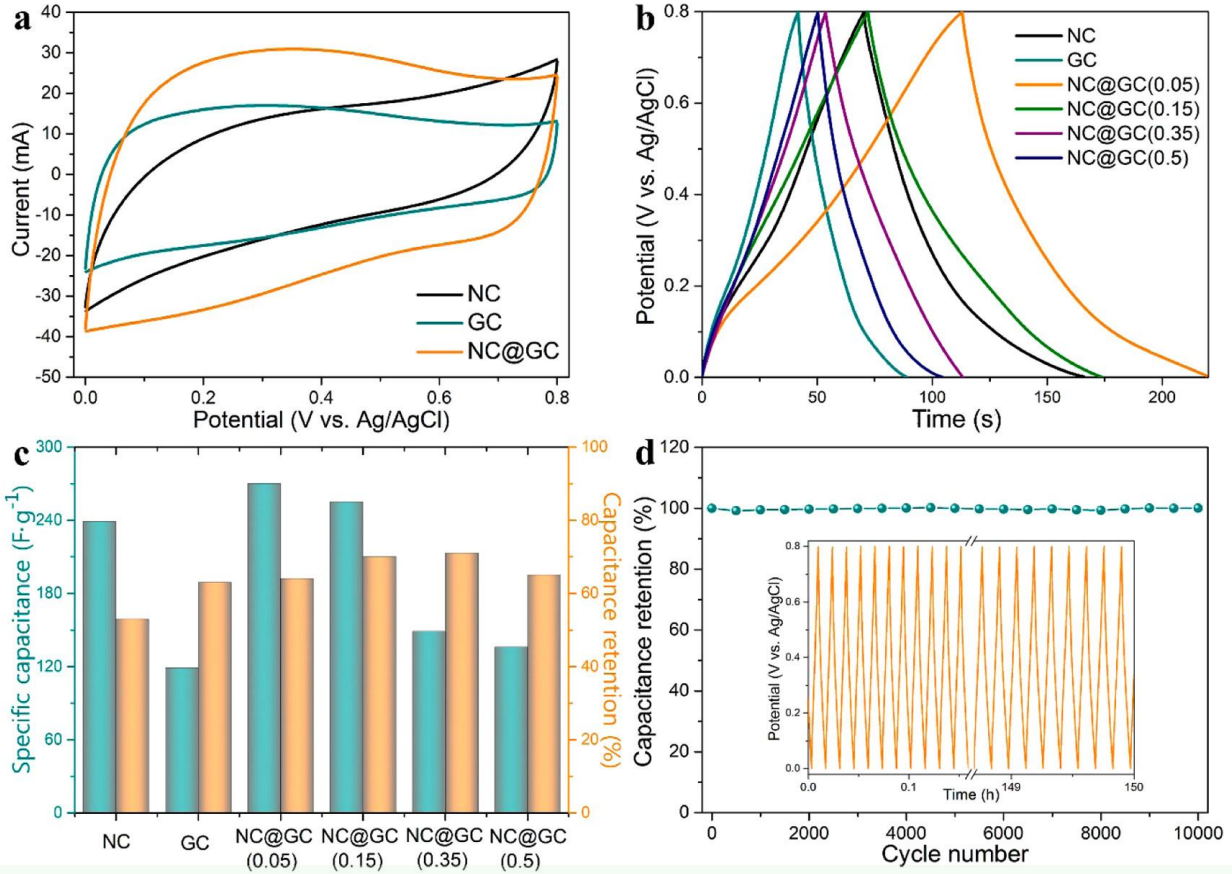


Thermal Conversion of Core–Shell Metal–Organic Frameworks: A New Method for Selectively Functionalized Nanoporous Hybrid Carbon | Journal of the American Chemical Society (acs.org)

Core-shell MOF for Supercapacitor







1. Core–shell ZIF-8@ZIF-67 crystals, which integrate the properties of single ZIF-8 and ZIF-67,
2. The core sizes of ZIF-8 and the shell thicknesses of ZIF-67 can be tuned simply by using different sizes of ZIF-8 seeds and varying the feeding molar ratios of Co^{2+}/Zn^{2+}.
3. After the direct carbonization of core–shell ZIF-8@ZIF-67 crystals →retain both the high surface area (up to 1276 $m^2 \cdot g^{-1}$) and high N content (10.6 wt %) derived from core ZIF-8 and superior graphitic structure originated from shell ZIF-67.
4. The optimized NC@GC exhibits a distinguished electrochemical performance as an electrode in supercapacitors.

Thermal Conversion of Core–Shell Metal–Organic Frameworks: A New Method for Selectively Functionalized Nanoporous Hybrid Carbon | Journal of the American Chemical Society (acs.org)

Nanomaterials

Students

https://go.blueja.io/6w-6OubAW0-8vmUcmyfrFQ

To access the evaluation, scan this QR code with your mobile phone.